

TrES-1b

R-0198

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_230709 · created 2026-07-16T21:56:18+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460134.7584 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1657 +/- 0.0083
Transit depth (Rp/Rs)^2	2.74 +/- 0.28 [%]
Orbital Inclination (inc)	89.84 +/- 0.69
Airmass coefficient 1 (a1)	1.0344 +/- 0.0095
Airmass coefficient 2 (a2)	-0.031 +/- 0.016
Scatter in the residuals of the lightcurve fit is	0.92 %
Best Comparison Star	#2 - [447, 393]
Optimal Method	PSF photometry
Transit Duration (day)	0.1065 +/- 0.0012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.1657 ± 0.0083 vs 0.1358 (deviation 3.6σ)
Duration fit vs published	0.1065 d vs 0.1042 d (deviation 1.94σ)
Mid-time O–C	1.7 min
Depth SNR	20.0
Residual scatter	0.92 %

Light curve

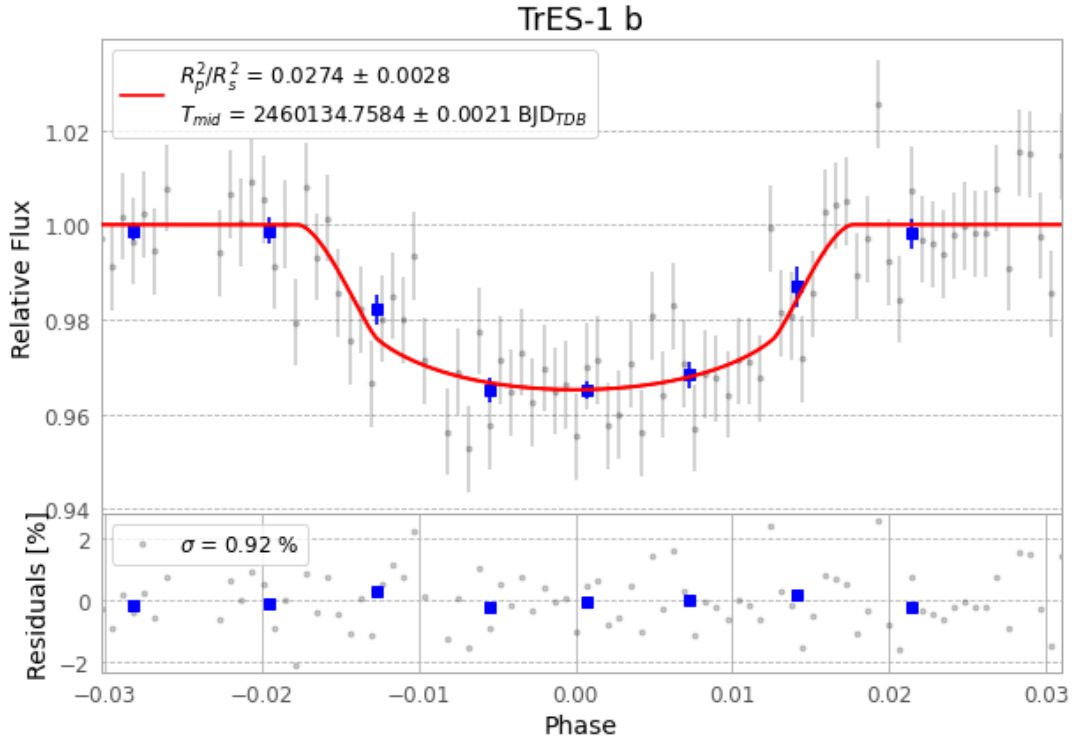


Figure 1 — FinalLightCurve_TrES-1 b_2023-07-08.png (SHA-256 0f348d7e16c17445...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [272, 214], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 512b42e3c3c7ced9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1965f5e

Data provenance

90 FITS frames (59 MB), TRES-1230709035415.FITS → TRES-1230709082115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-230709.log (c3f29a73ee78...), FinalLightCurve_TrES-1 b_2023-07-08.png (0f348d7e16c1...), FinalLightCurve_TrES-1 b_2023-07-08.csv (e2836b4643f6...)

Compute resources

Wall time 175.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 21:56Z.